

TypeScript & Playwright Workshop

Three-Day Outline

Day 1: TypeScript Fundamentals

Target: *Be comfortable reading and writing the TypeScript you'll see in Playwright tests.*

Topics Covered

- What is TypeScript, and why use it
- Setting up a TypeScript project
- Basic type annotations
- Writing and typing functions
- Union types
- Using any and unknown safely
- Interfaces for objects
- Structural typing
- Destructuring
- Async/await
- Enums

Lab

Hands-on practice writing TypeScript interfaces, typed arrays, and functions with typed parameters and return values.

Day 2: Introduction to Playwright

Target: *Understand how a Playwright test is built and walk through real examples together.*

Topics Covered

- Review of Day 1 lab
- What is Playwright, and why use it for test automation
- Setting up a Playwright project
- Anatomy of a Playwright test
- Locators and selectors
- Common actions (click, fill, navigate, etc.)
- Assertions
- Working through example test scenarios on a sample web application
- Introduction to test fixtures

Looking Ahead

Today's examples are designed to prepare you for tomorrow's independent lab. Pay close attention to the patterns used in each scenario.

Day 3: Lab & Advanced Scenarios

Target: Write your own Playwright tests independently, then review and discuss as a group.

Topics Covered

- Independent lab: writing your own Playwright test scenarios
- More advanced Playwright scenarios and techniques
- Group review and discussion of lab work
- Wrap-up and next steps

Lab

Using the sample application and patterns from Day 2, write your own Playwright test scenarios. We'll review selected examples together at the end of the day.

Technical Requirements

Please have the following installed and working before Day 1:

- Node.js (includes npm) — npm is what we'll use to install TypeScript and Playwright
- A code editor — Visual Studio Code is recommended
- Git — needed to clone the class repository

Note: On Day 2, installing Playwright will also download browser binaries (Chromium, Firefox, and WebKit), which can take a while and may be blocked on locked-down corporate machines. If you're on a work laptop with restricted software installs, please confirm with your IT team beforehand that this won't be an issue.

Materials

- [GitHub repository](#) with day-by-day code-alongs and the example front-end application: